MATHS MANTHAN

• TUITION CLASSES •

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Subject: Mathematics | **CBSE CLASS 10 – PERIODIC TEST 13 (SOLUTIONS)** | **Time:** 60 Minutes

Class: Class X (CBSE) | **Chapter 5: Arithmetic Progressions** | **MM: 30**

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CBSE STEP-MARKING GUIDELINES & EVALUATION PROTOCOL:

1. Full marks require explicit formula declaration, all intermediate algebraic steps, and clearly boxed final answers.
2. In internal choice (**OR**) questions, both options are solved completely with identical step-marking rigor.
3. Partial marks are allocated per step as indicated in brackets: **[0.5M, 1M, 1.5M, 2M, 3M]**.

Official Evaluation Key | Maths Manthan Meerut

SECTION A: Multiple Choice & Assertion-Reason MATHS MANTHAN | (5 × 1 = 5 Marks)

Q1. In an AP, if the common difference $d = 3$, then $a_{20} - a_{15}$ equals: [1]

(A) 5 (B) 15 (C) 8 (D) 12

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 1 Mark]

Correct Answer: **(B) 15** [0.5 Mark]

We apply the term-difference property:

Formula / Theorem Used: $a_m - a_n = (m - n)d$

Justification: [0.5 Mark]

$a_{20} - a_{15} = (a + 19d) - (a + 14d) = 5d = 5(3) = \mathbf{15}$.

Q2. The nth term of the AP $\frac{1}{m}, \frac{1+m}{m}, \frac{1+2m}{m}, \ldots$ $(m \neq 0)$ is: [1]

(A) $\frac{1+nm}{m}$ (B) $\frac{1+(n-1)m}{m}$ (C) $\frac{n+m}{m}$ (D) $\frac{m+(n-1)}{m}$

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 1 Mark]

Correct Answer: **(B)** $\boldsymbol{\frac{1+(n-1)m}{m}}$ [0.5 Mark]

Given: $a = \frac{1}{m}$ and $d = \frac{1+m}{m} - \frac{1}{m} = \frac{m}{m} = 1$.

Formula / Theorem Used: $a_n = a + (n-1)d$

Justification: [0.5 Mark]

$a_n = \frac{1}{m} + (n-1)(1) = \frac{\mathbf{1+(n-1)m}}{\mathbf{m}}$.

Q3. How many two-digit numbers are divisible by 3? [1]

(A) 25 (B) 30 (C) 32 (D) 33

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 1 Mark]

Correct Answer: **(B) 30** [0.5 Mark]

Given: AP is $12, 15, 18, \ldots, 99$ ($a = 12, d = 3, l = 99$).

Formula / Theorem Used: $l = a + (n-1)d$

Justification: [0.5 Mark]

$12 + 3(n-1) = 99 \implies 3(n-1) = 87 \implies n - 1 = 29 \implies n = \mathbf{30}$.

Q4. In an AP, if $a_{21} - a_7 = 84$, then its common difference d is: [1]

(A) 4 (B) 5 (C) 6 (D) 7

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 1 Mark]

Correct Answer: **(C) 6** [0.5 Mark]

Using $a_n = a + (n-1)d$:

Formula / Theorem Used: $a_{21} - a_7 = (a + 20d) - (a + 6d) = 14d$

Justification: [0.5 Mark]

$14d = 84 \implies d = \frac{84}{14} = \mathbf{6}$.

Q5. **Assertion (A):** The sum of the first n terms of an AP is $S_n = \frac{n}{2}(a + l)$.

Reason (R): Substituting $l = a + (n-1)d$ into $S_n = \frac{n}{2}(a + l)$ yields $S_n = \frac{n}{2}[2a + (n-1)d]$. [1]

(A) Both (A) and (R) are true, and (R) is correct explanation.

(B) Both (A) and (R) are true, and (R) is not correct explanation.

(C) Assertion (A) is true, but Reason (R) is false.

(D) Assertion (A) is false, but Reason (R) is true.

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 1 Mark]

Correct Answer: **(A) Both (A) and (R) are true, and (R) explains (A).** [0.5 Mark]

Connecting the two standard forms of S_n:

Formula / Theorem Used: $S_n = \frac{n}{2}[a + (a + (n-1)d)] = \frac{n}{2}[2a + (n-1)d]$

Justification: [0.5 Mark]

Assertion (A) is true, and Reason (R) states the exact algebraic equivalence connecting l and d (true), supporting and explaining (A).

Q6. In an AP, $a = 2$, $d = 8$, and $S_n = 90$. Find n and a_n. **[2]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Given: $a = 2$, $d = 8$, $S_n = 90$.

Formula / Theorem Used: $S_n = \frac{n}{2}[2a + (n-1)d]$ and $a_n = a + (n-1)d$

Step 1: Solving Quadratic for n [1 Mark]

$$\frac{n}{2}[2(2) + (n-1)(8)] = 90 \implies \frac{n}{2}(8n - 4) = 90 \implies n(4n - 2) = 90 \implies 2n^2 - n - 45 = 0$$

Splitting middle term $(-10 + 9 = -1,\ (-10)(9) = -90)$: $(n-5)(2n+9) = 0 \implies n = 5$ (reject $-\frac{9}{2}$).

Step 2: Finding a_5 [1 Mark]

$a_5 = 2 + (5-1)(8) = 2 + 32 = 34 \implies$ $\boxed{\mathbf{n = 5,\quad a_5 = 34}}$.

Q7. Find the 10th term from the end of the AP $8, 10, 12, \ldots, 126$. **[2]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Step 1: Identifying Parameters [1 Mark]

Here $a = 8$, $d = 10 - 8 = 2$, and last term $l = 126$.

Formula / Theorem Used: n^{th} term from end $= l - (n-1)d$

Step 2: Computing 10th Term from End [1 Mark]

$$10^{\text{th}} \text{ term from end} = 126 - (10-1)(2) = 126 - 9 \times 2 = 126 - 18 = \boxed{\mathbf{108}}$$

Q8. If the 9th term of an AP is zero ($a_9 = 0$), prove that its 29th term is twice its 19th term. **[3]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Step 1: Expressing a in Terms of d [1 Mark]

Formula / Theorem Used: $a_n = a + (n-1)d$

$a_9 = a + 8d = 0 \implies a = -8d.$

Step 2: Computing a_{29} and a_{19} [1.5 Marks]

$a_{29} = a + 28d = -8d + 28d = 20d.$

$a_{19} = a + 18d = -8d + 18d = 10d.$

Step 3: Comparing Both Terms [0.5 Mark]

$a_{29} = 20d = 2(10d) =$ **$2a_{19}$** [Hence Proved].

Q9. If the 19th term of an AP is equal to three times its 6th term and its 9th term is 19, find the AP. **[3]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Given: $a_{19} = 3a_6$ and $a_9 = 19$.

Formula / Theorem Used: $a_n = a + (n-1)d$

Step 1: Setting Up Both Linear Equations [1.5 Marks]

$a + 18d = 3(a + 5d) \implies a + 18d = 3a + 15d \implies 2a = 3d \implies a = \frac{3}{2}d.$ — (i)

$a_9 = a + 8d = 19.$ — (ii)

Step 2: Solving for d, a, and Writing the AP [1.5 Marks]

Substituting $a = \frac{3}{2}d$ into (ii): $\frac{3}{2}d + 8d = 19 \implies \frac{3d+16d}{2} = 19 \implies \frac{19d}{2} = 19 \implies d = 2.$

Thus $a = \frac{3}{2}(2) = 3.$

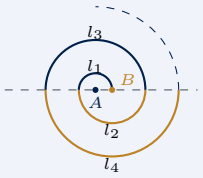
Required AP: **3**, **5**, **7**, **9**, ...

Q10. A spiral is made up of successive semicircles, with centres alternately at A and B, starting with centre at A, of radii 0.5 cm, 1.0 cm, 1.5 cm, 2.0 cm, ... What is the total length of such a spiral made up of thirteen consecutive semicircles? (Take $\pi = \frac{22}{7}$.) **[5]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 5 Marks]

Step 1: Semicircular Lengths as an AP [1.5 Marks]

Radii are $r_1 = 0.5$ cm, $r_2 = 1.0$ cm, ..., $r_{13} = 6.5$ cm.
Arc length $l_k = \pi r_k$ gives the AP: $\frac{\pi}{2}$, π, $\frac{3\pi}{2}$, ..., $\frac{13\pi}{2}$.



Formula / Theorem Used: $S_n = \frac{n}{2}(a + l)$ where $l_k = \pi r_k$

Step 2: Identifying Parameters (a, d, l_{13}) [1.5 Marks]

First arc $a = l_1 = \frac{\pi}{2}$, common difference $d = \pi - \frac{\pi}{2} = \frac{\pi}{2}$, last arc $l_{13} = \pi(13 \times 0.5) = \frac{13\pi}{2}$.

Step 3: Summing 13 Semicircles [2 Marks]

$$S_{13} = l_1 + l_2 + \cdots + l_{13} = \frac{13}{2}\left(\frac{\pi}{2} + \frac{13\pi}{2}\right) = \frac{13}{2}(7\pi) = \frac{13}{2} \times 7 \times \frac{22}{7} = 13 \times 11 = \boxed{\mathbf{143}\text{ cm}}$$

Q11. If the sum of the first n terms of an AP is $S_n = 3n^2 + 5n$ and its mth term is 164, find the value of m. Also find its 20th term. **[5]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 5 Marks]

Given: $S_n = 3n^2 + 5n$ and $a_m = 164$.

Formula / Theorem Used: $a_1 = S_1$, $a_n = S_n - S_{n-1}$, $a_m = a + (m-1)d$

Step 1: Finding a and d (or General Term a_n) [2 Marks]

$a_1 = S_1 = 3(1)^2 + 5(1) = 8$. $S_2 = 3(2)^2 + 5(2) = 12 + 10 = 22 \implies a_2 = 22 - 8 = 14$.
Hence first term $a = 8$, common difference $d = 14 - 8 = 6$, and $a_n = 8 + (n-1)(6) = 6n + 2$.

Step 2: Solving for m from $a_m = 164$ [1.5 Marks]

$$6m + 2 = 164 \implies 6m = 162 \implies m = \frac{162}{6} = 27$$

Step 3: Calculating the 20th Term (a_{20}) [1.5 Marks]

$$a_{20} = 6(20) + 2 = 120 + 2 = 122 \implies \boxed{\mathbf{m = 27}, \quad \mathbf{a_{20} = 122}}$$

Q12. CASE STUDY: Career Increments (Subba Rao's Employment Schedule)

Subba Rao started work in the year 1995 at an annual salary of Rs. 5,000 and received an annual increment of Rs. 200 each year thereafter. [5]

(i) Write the AP of his annual salaries and identify a and d. [1]

(ii) What was Subba Rao's annual salary in the year 2005? [1]

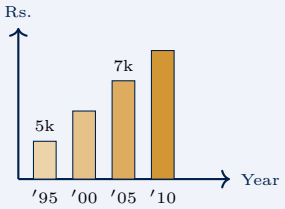
(iii) In which calendar year did his annual salary reach Rs. 7,000? [3]

— **OR** —

Calculate the total cumulative salary earned by Subba Rao in his first 15 years of service. [3]

▶ **CASE-BASED SOLUTION & MARKING SCHEME:** [Total: 5 Marks]

Extracted Scenario Data: Year 1 corresponds to 1995 ($a_1 = 5{,}000$), with annual increment $d = 200$.



Formula / Theorem Used: $a_n = a + (n-1)d$ and $S_n = \frac{n}{2}[2a + (n-1)d]$

Sub-part (i): Salary AP, a, and d [1 Mark]

Annual salaries: **5,000, 5,200, 5,400, ...** with **a = Rs. 5,000** and **d = Rs. 200**.

Sub-part (ii): Annual Salary in 2005 [1 Mark]

Since 1995 is Year 1, 2005 is the $(2005 - 1995 + 1) = 11^{\text{th}}$ year of service ($n = 11$):

$$a_{11} = 5{,}000 + (11-1)(200) = 5{,}000 + 2{,}000 = \boxed{\text{Rs. } \mathbf{7{,}000}}$$

Sub-part (iii): Year When Salary Reached Rs. 7,000 [3 Marks]

Setting $a_n = 7{,}000$:

$$5{,}000 + (n-1)(200) = 7{,}000 \implies 200(n-1) = 2{,}000 \implies n - 1 = 10 \implies n = 11^{\text{th}} \text{ year}$$

Calendar year $= 1995 + (11 - 1) =$ **2005**.

Sub-part (iii) [OR Option]: Total Salary Earned in First 15 Years [3 Marks]

$$S_{15} = \frac{15}{2}[2(5{,}000) + (15-1)(200)] = \frac{15}{2}(10{,}000 + 2{,}800) = \frac{15}{2} \times 12{,}800 = 15 \times 6{,}400 = \boxed{\text{Rs. } \mathbf{96{,}000}}$$

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